AFDX 송신 지터 종합 리포트

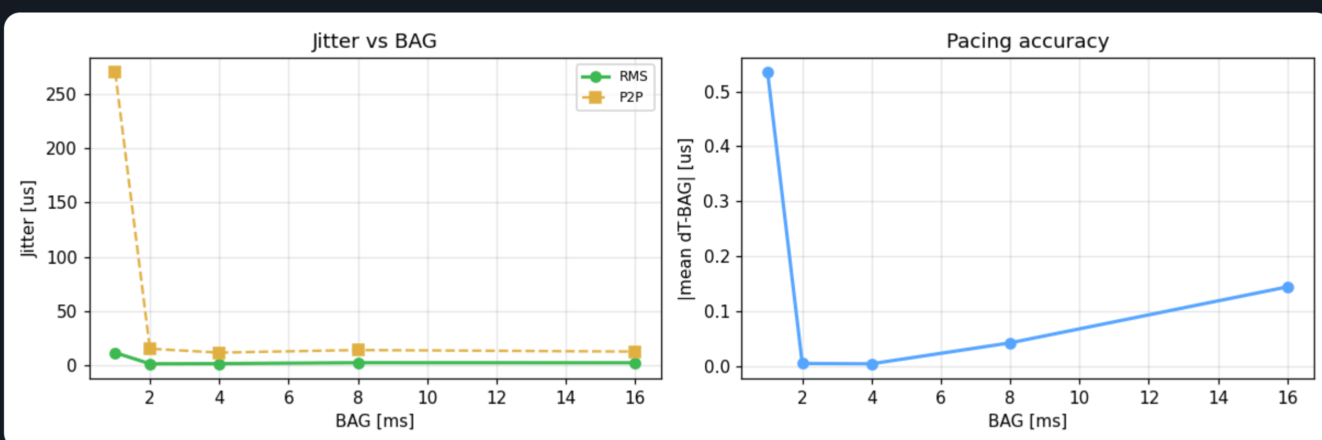
KFDX(FPGA) → PC enp4s0(igc) 외부 캡처 · `J(n)=ΔTtx−BAG` · 시스템 격리(measure_setup.sh) 적용 · 5개 BAG

요약

BAG(×10MS)	목표	평균ΔT(MS)	오차(MS)	MIN	MAX	JITTER RMS	|J|MAX	P2P	표본
100	1.000ms	1000.53	+0.53	994.2	1263.9	**11.61**	263.9	269.7	592
200	2.000ms	2000.00	-0.00	1992.9	2008.2	**1.42**	8.2	15.3	460
400	4.000ms	4000.00	-0.00	3993.5	4005.2	**1.57**	6.5	11.7	326
800	8.000ms	8000.04	+0.04	7992.3	8006.3	**2.39**	7.7	14.1	189
1600	16.000ms	16000.14	+0.14	15994.5	16007.2	**2.34**	7.2	12.6	126

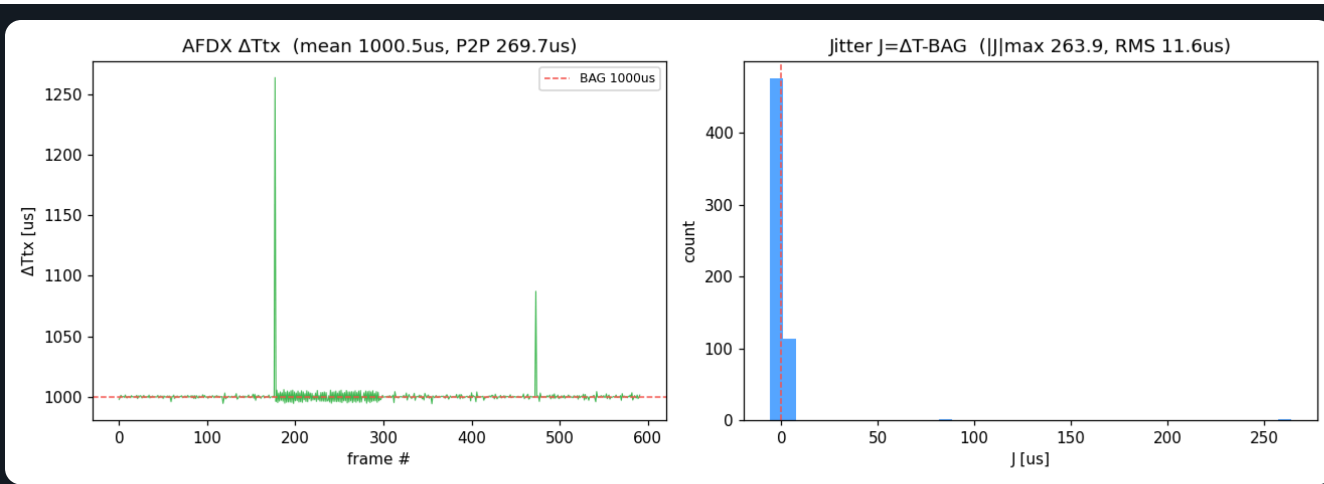
· 평균 ΔTtx가 BAG와 일치 = FPGA 페이싱 정확 · Jitter RMS = 송출 간격 변동(작을수록 좋음)

추세



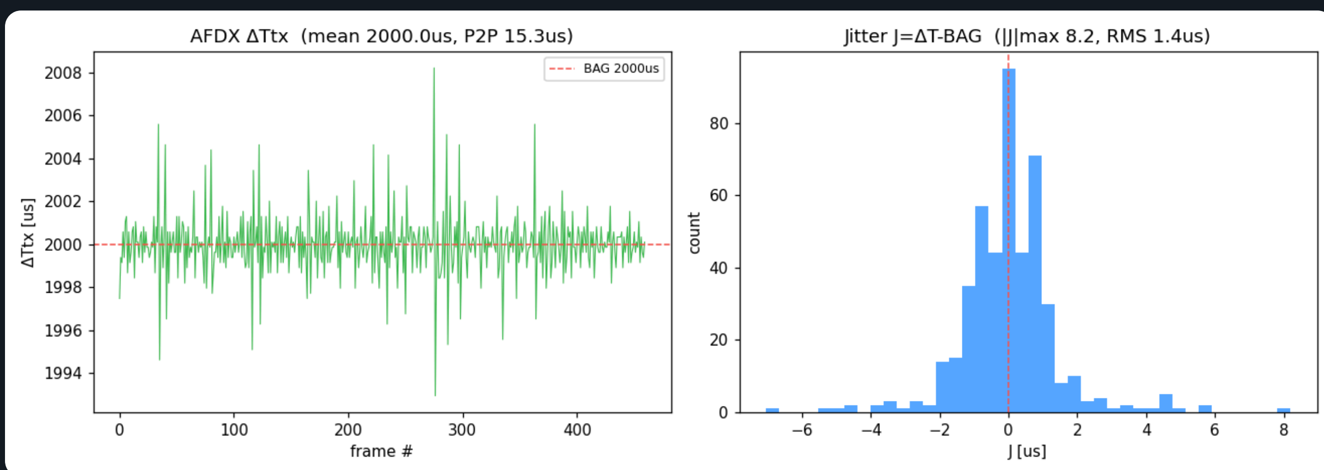
BAG 100 · 1.000ms

평균 ΔTtx **1000.53μs** · Jitter RMS **11.61μs** · |J|max 263.9μs · P2P 269.7μs · 표본 592



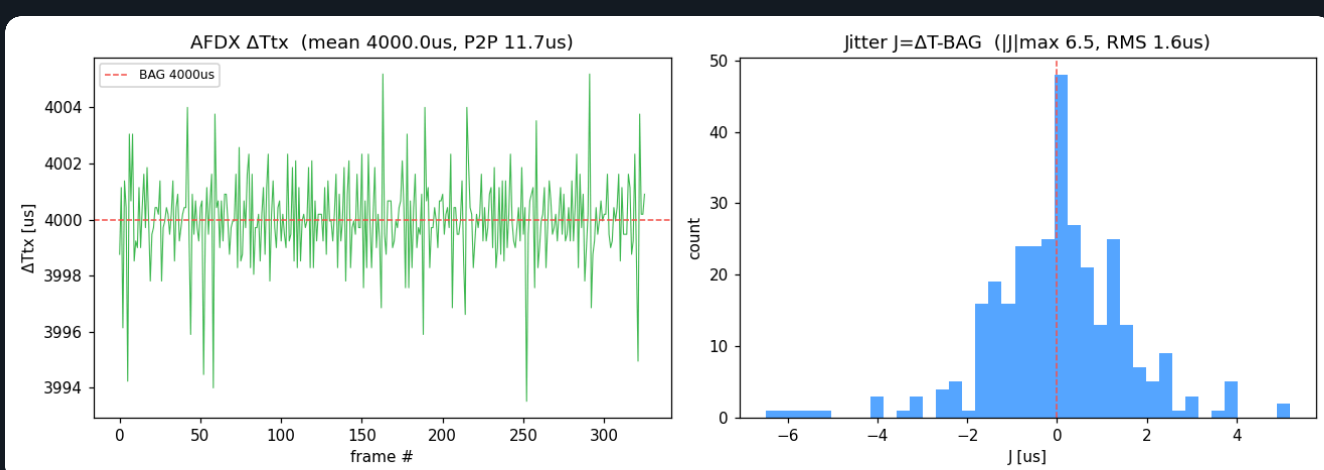
BAG 200 · 2.000ms

평균 ΔTtx **2000.00μs** · Jitter RMS **1.42μs** · |J|max 8.2μs · P2P 15.3μs · 표본 460



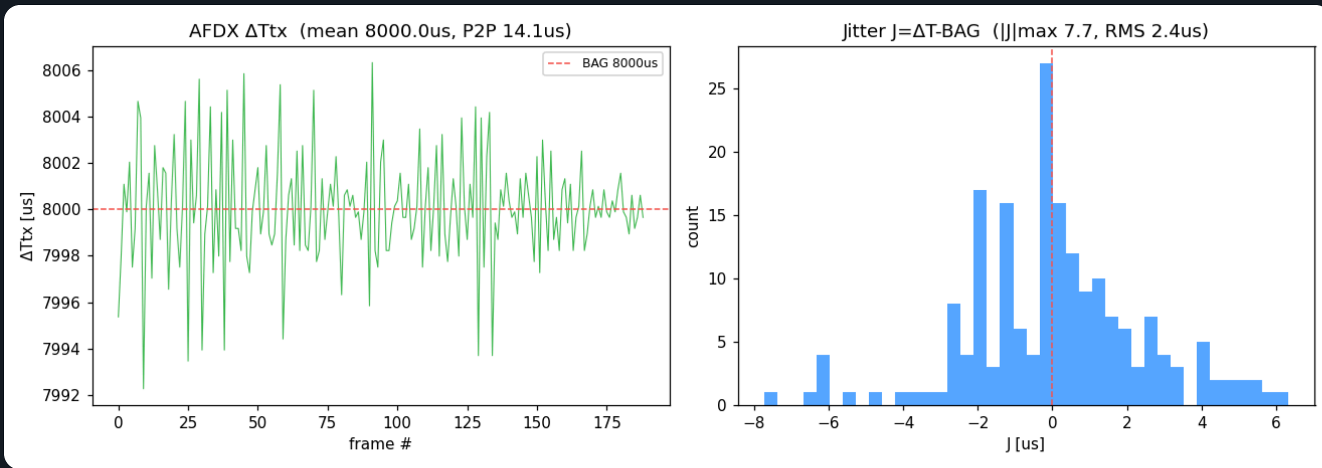
BAG 400 · 4.000ms

평균 ΔTtx **4000.00μs** · Jitter RMS **1.57μs** · |J|max 6.5μs · P2P 11.7μs · 표본 326



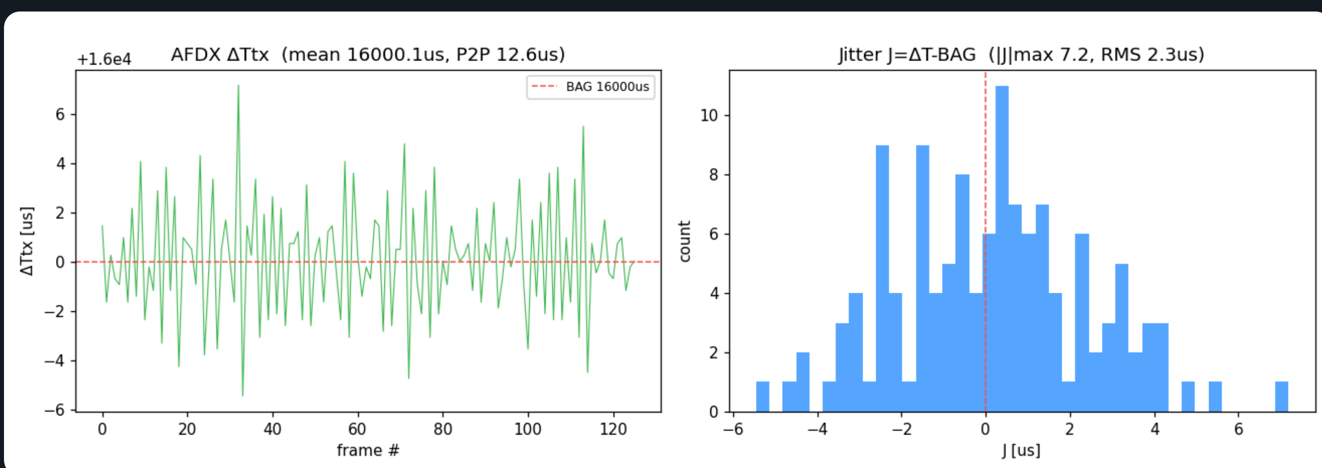
BAG 800 · 8.000ms

평균 ΔTtx **8000.04μs** · Jitter RMS **2.39μs** · |J|max 7.7μs · P2P 14.1μs · 표본 189



BAG 1600 · 16.000ms

평균 ΔTtx **16000.14μs** · Jitter RMS **2.34μs** · |J|max 7.2μs · P2P 12.6μs · 표본 126



생성: report.py · 데이터 results/rep_*.json